

Category	Element	Symbol	AtomicMass	Numberoff	Numberoff	Numberoff	Period	Group	Phase
Con	Hydrogen	H	1.007	0	1	1	1	1	1 gas
Div	Helium	He	4.002	2	2	2	1	18	gas
Off	Lithium	Li	6.941	4	3	3	2	1	solid
Neu	Beryllium	Be	9.012	5	4	4	2	2	solid
Div	Boron	B	10.811	6	5	5	2	13	solid
Con	Carbon	C	12.011	6	6	6	2	14	solid
Off	Nitrogen	N	14.007	7	7	7	2	15	gas
Ret	Oxygen	O	15.999	8	8	8	2	16	gas
Div	Fluorine	F	18.998	10	9	9	2	17	gas
Div	Neon	Ne	20.18	10	10	10	2	18	gas
Neu	Sodium	Na	22.99	12	11	11	3	1	solid
Neu	Magnesium	Mg	24.305	12	12	12	3	2	solid
Off	Aluminum	Al	26.982	14	13	13	3	13	solid
Div	Silicon	Si	28.086	14	14	14	3	14	solid
Off	Phosphorus	P	30.974	16	15	15	3	15	solid
Con	Sulfur	S	32.065	16	16	16	3	16	solid
Div	Chlorine	Cl	35.453	18	17	17	3	17	gas
Neu	Argon	Ar	39.948	22	18	18	3	18	gas
Off	Potassium	K	39.098	20	19	19	4	1	solid
Neu	Calcium	Ca	40.078	20	20	20	4	2	solid
Off	Scandium	Sc	44.956	24	21	21	4	3	solid
Con	Titanium	Ti	47.867	26	22	22	4	4	solid
Off	Vanadium	V	50.942	28	23	23	4	5	solid
Con	Chromium	Cr	51.996	28	24	24	4	6	solid
Off	Manganese	Mn	54.938	30	25	25	4	7	solid
Con	Iron	Fe	55.845	30	26	26	4	8	solid
Off	Cobalt	Co	58.933	32	27	27	4	9	solid
Neu	Nickel	Ni	58.693	31	28	28	4	10	solid
Div	Copper	Cu	63.546	35	29	29	4	11	solid
Off	Zinc	Zn	65.38	35	30	30	4	12	solid
Off	Gallium	Ga	69.723	39	31	31	4	13	solid
Div	Germanium	Ge	72.64	41	32	32	4	14	solid
Off	Arsenic	As	74.922	42	33	33	4	15	solid
Ret	Selenium	Se	78.96	45	34	34	4	16	solid
Div	Bromine	Br	79.904	45	35	35	4	17	liq
Div	Krypton	Kr	83.798	48	36	36	4	18	gas
Off	Rubidium	Rb	85.468	48	37	37	5	1	solid
Neu	Strontium	Sr	87.62	50	38	38	5	2	solid
Off	Yttrium	Y	88.906	50	39	39	5	3	solid
Con	Zirconium	Zr	91.224	51	40	40	5	4	solid
Off	Niobium	Nb	92.906	52	41	41	5	5	solid
Con	Molybdenum	Mo	95.96	54	42	42	5	6	solid
Anom	Technetium	Tc	98	55	43	43	5	7	artificial

Con	Ruthenium	Ru	101.07	57	44	44	5	8 solid
Off	Rhodium	Rh	102.906	58	45	45	5	9 solid
Con	Palladium	Pd	106.42	60	46	46	5	10 solid
Con	Silver	Ag	107.868	61	47	47	5	11 solid
Off	Cadmium	Cd	112.411	64	48	48	5	12 solid
Off	Indium	In	114.818	66	49	49	5	13 solid
Ret	Tin	Sn	118.71	69	50	50	5	14 solid
Off	Antimony	Sb	121.76	71	51	51	5	15 solid
Ret	Tellurium	Te	127.6	76	52	52	5	16 solid
Ret	Iodine	I	126.904	74	53	53	5	17 solid
Div	Xenon	Xe	131.293	77	54	54	5	18 gas
Off	Cesium	Cs	132.905	78	55	55	6	1 solid
Neu	Barium	Ba	137.327	81	56	56	6	2 solid
Off	Lanthanum	La	138.905	82	57	57	6	3 solid
Ret	Cerium	Ce	140.116	82	58	58	6	solid
Off	Praseodymium	Pr	140.908	82	59	59	6	solid
Off	Neodymium	Nd	144.242	84	60	60	6	solid
Anom	Promethium	Pm	145	84	61	61	6	artificial
Ret	Samarium	Sm	150.36	88	62	62	6	solid
Ret	Europium	Eu	151.964	89	63	63	6	solid
Off	Gadolinium	Gd	157.25	93	64	64	6	solid
Off	Terbium	Tb	158.925	94	65	65	6	solid
Ret	Dysprosium	Dy	162.5	97	66	66	6	solid
Off	Holmium	Ho	164.93	98	67	67	6	solid
Ret	Erbium	Er	167.259	99	68	68	6	solid
Off	Thulium	Tm	168.934	100	69	69	6	solid
Neu	Ytterbium	Yb	173.054	103	70	70	6	solid
Off	Lutetium	Lu	174.967	104	71	71	6	solid
Con	Hafnium	Hf	178.49	106	72	72	6	4 solid
Off	Tantalum	Ta	180.948	108	73	73	6	5 solid
Con	Wolfram	W	183.84	110	74	74	6	6 solid
Off	Rhenium	Re	186.207	111	75	75	6	7 solid
Con	Osmium	Os	190.23	114	76	76	6	8 solid
Con	Iridium	Ir	192.217	115	77	77	6	9 solid
Con	Platinum	Pt	195.084	117	78	78	6	10 solid
Con	Gold	Au	196.967	118	79	79	6	11 solid
Anom	Mercury	Hg	200.59	121	80	80	6	12 liq
Off	Thallium	Tl	204.383	123	81	81	6	13 solid
Anom	Lead	Pb	207.2	125	82	82	6	14 solid
Off	Bismuth	Bi	208.98	126	83	83	6	15 solid
Anom	Polonium	Po	210	126	84	84	6	16 solid
Anom	Astatine	At	210	125	85	85	6	17 solid
Div	Radon	Rn	222	136	86	86	6	18 gas
Anom	Francium	Fr	223	136	87	87	7	1 solid

Off	Radium	Ra	226	138	88	88	7	2 solid
Off	Actinium	Ac	227	138	89	89	7	3 solid
Off	Thorium	Th	232.038	142	90	90	7	solid
Off	Protactiniu	Pa	231.036	140	91	91	7	solid
Off	Uranium	U	238.029	146	92	92	7	solid
Anom	Neptuniu	Np	237	144	93	93	7	artificial
Anom	Plutonium	Pu	244	150	94	94	7	artificial
Anom	Americium	Am	243	148	95	95	7	artificial
Anom	Curium	Cm	247	151	96	96	7	artificial
Anom	Berkelium	Bk	247	150	97	97	7	artificial
Anom	Californiu	Cf	251	153	98	98	7	artificial
Anom	Einsteinium	Es	252	153	99	99	7	artificial
Anom	Fermium	Fm	257	157	100	100	7	artificial
Anom	Mendeleviu	Md	258	157	101	101	7	artificial
Off	Nobelium	No	259	157	102	102	7	artificial
Off	Lawrenciu	Lr	262	159	103	103	7	artificial
Off	Rutherford	Rf	261	157	104	104	7	4 artificial
Anom	Dubnium	Db	262	157	105	105	7	5 artificial
Off	Seaborgiu	Sg	266	160	106	106	7	6 artificial
Anom	Bohrium	Bh	264	157	107	107	7	7 artificial
Off	Hassium	Hs	267	159	108	108	7	8 artificial
Anom	Meitnerium	Mt	268	159	109	109	7	9 artificial
Anom	Darmstadt	Ds	271	161	110	110	7	10 artificial
Off	Roentgeniu	Rg	272	161	111	111	7	11 artificial
Anom	Coperniciu	Cn	285	173	112	112	7	12 artificial
Anom	Nihonium	Nh	284	171	113	113	7	13 artificial
Anom	Flerovium	Fl	289	175	114	114	7	14 artificial
Anom	Moscovium	Mc	288	173	115	115	7	15 artificial
Anom	Livermoriu	Lv	292	176	116	116	7	16 artificial
Anom	Tennessinu	Ts	295	178	117	117	7	17 artificial
Anom	Oganesson	Og	294	176	118	118	7	18 artificial

Radioactive	Natural	Metal	Nonmetal	Metalloid	Type	AtomicRad	Electroneg	FirstIoniz	Density
	yes		yes		Nonmetal	0.79	2.2	13.5984	8.99E-05
	yes		yes		Noble Gas	0.49		24.5874	1.79E-04
	yes	yes			Alkali Meta	2.1	0.98	5.3917	5.34E-01
	yes	yes			Alkaline Ea	1.4	1.57	9.3227	1.85E+00
	yes			yes	Metalloid	1.2	2.04	8.298	2.34E+00
	yes		yes		Nonmetal	0.91	2.55	11.2603	2.27E+00
	yes		yes		Nonmetal	0.75	3.04	14.5341	1.25E-03
	yes		yes		Nonmetal	0.65	3.44	13.6181	1.43E-03
	yes		yes		Halogen	0.57	3.98	17.4228	1.70E-03
	yes		yes		Noble Gas	0.51		21.5645	9.00E-04
	yes	yes			Alkali Meta	2.2	0.93	5.1391	9.71E-01
	yes	yes			Alkaline Ea	1.7	1.31	7.6462	1.74E+00
	yes	yes			Metal	1.8	1.61	5.9858	2.70E+00
	yes			yes	Metalloid	1.5	1.9	8.1517	2.33E+00
	yes		yes		Nonmetal	1.2	2.19	10.4867	1.82E+00
	yes		yes		Nonmetal	1.1	2.58	10.36	2.07E+00
	yes		yes		Halogen	0.97	3.16	12.9676	3.21E-03
	yes		yes		Noble Gas	0.88		15.7596	1.78E-03
	yes	yes			Alkali Meta	2.8	0.82	4.3407	8.62E-01
	yes	yes			Alkaline Ea	2.2	1	6.1132	1.54E+00
	yes	yes			Transition I	2.1	1.36	6.5615	2.99E+00
	yes	yes			Transition I	2	1.54	6.8281	4.54E+00
	yes	yes			Transition I	1.9	1.63	6.7462	6.11E+00
	yes	yes			Transition I	1.9	1.66	6.7665	7.15E+00
	yes	yes			Transition I	1.8	1.55	7.434	7.44E+00
	yes	yes			Transition I	1.7	1.83	7.9024	7.87E+00
	yes	yes			Transition I	1.7	1.88	7.881	8.86E+00
	yes	yes			Transition I	1.6	1.91	7.6398	8.91E+00
	yes	yes			Transition I	1.6	1.9	7.7264	8.96E+00
	yes	yes			Transition I	1.5	1.65	9.3942	7.13E+00
	yes	yes			Metal	1.8	1.81	5.9993	5.91E+00
	yes			yes	Metalloid	1.5	2.01	7.8994	5.32E+00
	yes			yes	Metalloid	1.3	2.18	9.7886	5.78E+00
	yes		yes		Nonmetal	1.2	2.55	9.7524	4.81E+00
	yes		yes		Halogen	1.1	2.96	11.8138	3.12E+00
	yes		yes		Noble Gas	1		13.9996	3.73E-03
	yes	yes			Alkali Meta	3	0.82	4.1771	1.53E+00
	yes	yes			Alkaline Ea	2.5	0.95	5.6949	2.64E+00
	yes	yes			Transition I	2.3	1.22	6.2173	4.47E+00
	yes	yes			Transition I	2.2	1.33	6.6339	6.51E+00
	yes	yes			Transition I	2.1	1.6	6.7589	8.57E+00
	yes	yes			Transition I	2	2.16	7.0924	1.02E+01
yes		yes			Transition I	2	1.9	7.28	1.15E+01

	yes	yes		Transition I	1.9	2.2	7.3605	1.24E+01
	yes	yes		Transition I	1.8	2.28	7.4589	1.24E+01
	yes	yes		Transition I	1.8	2.2	8.3369	1.20E+01
	yes	yes		Transition I	1.8	1.93	7.5762	1.05E+01
	yes	yes		Transition I	1.7	1.69	8.9938	8.69E+00
	yes	yes		Metal	2	1.78	5.7864	7.31E+00
	yes	yes		Metal	1.7	1.96	7.3439	7.29E+00
	yes		yes	Metalloid	1.5	2.05	8.6084	6.69E+00
	yes		yes	Metalloid	1.4	2.1	9.0096	6.23E+00
	yes	yes		Halogen	1.3	2.66	10.4513	4.93E+00
	yes	yes		Noble Gas	1.2		12.1298	5.89E-03
	yes	yes		Alkali Meta	3.3	0.79	3.8939	1.87E+00
	yes	yes		Alkaline Ea	2.8	0.89	5.2117	3.59E+00
	yes	yes		Lanthanide	2.7	1.1	5.5769	6.15E+00
	yes	yes		Lanthanide	2.7	1.12	5.5387	6.77E+00
	yes	yes		Lanthanide	2.7	1.13	5.473	6.77E+00
	yes	yes		Lanthanide	2.6	1.14	5.525	7.01E+00
yes		yes		Lanthanide	2.6	1.13	5.582	7.26E+00
	yes	yes		Lanthanide	2.6	1.17	5.6437	7.52E+00
	yes	yes		Lanthanide	2.6	1.2	5.6704	5.24E+00
	yes	yes		Lanthanide	2.5	1.2	6.1501	7.90E+00
	yes	yes		Lanthanide	2.5	1.2	5.8638	8.23E+00
	yes	yes		Lanthanide	2.5	1.22	5.9389	8.55E+00
	yes	yes		Lanthanide	2.5	1.23	6.0215	8.80E+00
	yes	yes		Lanthanide	2.5	1.24	6.1077	9.07E+00
	yes	yes		Lanthanide	2.4	1.25	6.1843	9.32E+00
	yes	yes		Lanthanide	2.4	1.1	6.2542	6.97E+00
	yes	yes		Lanthanide	2.3	1.27	5.4259	9.84E+00
	yes	yes		Transition I	2.2	1.3	6.8251	1.33E+01
	yes	yes		Transition I	2.1	1.5	7.5496	1.67E+01
	yes	yes		Transition I	2	2.36	7.864	1.93E+01
	yes	yes		Transition I	2	1.9	7.8335	2.10E+01
	yes	yes		Transition I	1.9	2.2	8.4382	2.26E+01
	yes	yes		Transition I	1.9	2.2	8.967	2.26E+01
	yes	yes		Transition I	1.8	2.28	8.9587	2.15E+01
	yes	yes		Transition I	1.8	2.54	9.2255	1.93E+01
	yes	yes		Transition I	1.8	2	10.4375	1.35E+01
	yes	yes		Metal	2.1	2.04	6.1082	1.19E+01
	yes	yes		Metal	1.8	2.33	7.4167	1.13E+01
	yes	yes		Metal	1.6	2.02	7.2856	9.81E+00
yes	yes		yes	Metalloid	1.5	2	8.417	9.32E+00
yes	yes		yes	Noble Gas	1.4	2.2	9.3	7.00E+00
yes	yes	yes		Alkali Meta	1.3		10.7485	9.73E-03
yes	yes	yes		Alkaline Earth Metal		0.7	4.0727	1.87E+00

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14.175	20.28	3	14.304	1	1
	4.22	5	5.193	1	
453.85	1615	5	3.582	2	1
1560.15	2742	6	1.825	2	2
2573.15	4200	6	1.026	2	3
3948.15	4300	7	0.709	2	4
63.29	77.36	8	1.04	2	5
50.5	90.2	8	0.918	2	6
53.63	85.03	6	0.824	2	7
24.703	27.07	8	1.03	2	8
371.15	1156	7	1.228	3	1
923.15	1363	8	1.023	3	2
933.4	2792	8	0.897	3	3
1683.15	3538	8	0.705	3	4
317.25	553	7	0.769	3	5
388.51	717.8	10	0.71	3	6
172.31	239.11	11	0.479	3	7
83.96	87.3	8	0.52	3	8
336.5	1032	10	0.757	4	1
1112.15	1757	14	0.647	4	2
1812.15	3109	15	0.568	4	
1933.15	3560	9	0.523	4	
2175.15	3680	9	0.489	4	
2130.15	2944	9	0.449	4	
1519.15	2334	11	0.479	4	
1808.15	3134	10	0.449	4	
1768.15	3200	14	0.421	4	
1726.15	3186	11	0.444	4	
1357.75	2835	11	0.385	4	
692.88	1180	15	0.388	4	
302.91	2477	14	0.371	4	3
1211.45	3106	17	0.32	4	4
1090.15	887	14	0.329	4	5
494.15	958	20	0.321	4	6
266.05	332	19	0.474	4	7
115.93	119.93	23	0.248	4	8
312.79	961	20	0.363	5	1
1042.15	1655	18	0.301	5	2
1799.15	3609	21	0.298	5	
2125.15	4682	20	0.278	5	
2741.15	5017	24	0.265	5	
2890.15	4912	20	0.251	5	
2473.15	5150	23		5	

2523.15	4423	16	0.238	5	
2239.15	3968	20	0.243	5	
1825.15	3236	21	0.244	5	
1234.15	2435	27	0.235	5	
594.33	1040	22	0.232	5	
429.91	2345	34	0.233	5	3
505.21	2875	28	0.228	5	4
904.05	1860	29	0.207	5	5
722.8	1261	29	0.202	5	6
386.65	457.4	24	0.214	5	7
161.45	165.03	31	0.158	5	8
301.7	944	22	0.242	6	1
1002.15	2170	25	0.204	6	2
1193.15	3737	19	0.195	6	
1071.15	3716	19	0.192	6	
1204.15	3793	15	0.193	6	
1289.15	3347	16	0.19	6	
1204.15	3273	14		6	
1345.15	2067	17	0.197	6	
1095.15	1802	21	0.182	6	
1585.15	3546	17	0.236	6	
1630.15	3503	24	0.182	6	
1680.15	2840	21	0.17	6	
1743.15	2993	29	0.165	6	
1795.15	3503	16	0.168	6	
1818.15	2223	18	0.16	6	
1097.15	1469	16	0.155	6	
1936.15	3675	22	0.154	6	
2500.15	4876	17	0.144	6	
3269.15	5731	19	0.14	6	
3680.15	5828	22	0.132	6	
3453.15	5869	21	0.137	6	
3300.15	5285	19	0.13	6	
2716.15	4701	25	0.131	6	
2045.15	4098	32	0.133	6	
1337.73	3129	21	0.129	6	
234.43	630	26	0.14	6	
577.15	1746	28	0.129	6	3
600.75	2022	29	0.129	6	4
544.67	1837	19	0.122	6	5
527.15	1235	34		6	6
575.15	610	21		6	7
202.15	211.3	20	0.094	6	8
300.15	950	21		7	1

973.15	2010	15		7	2
1323.15	3471	11	0.12	7	
2028.15	5061	12	0.113	7	
1873.15	4300	14		7	
1405.15	4404	15	0.116	7	
913.15	4273	153		7	
913.15	3501	163		7	
1267.15	2880	133		7	
1340.15	3383	133		7	
1259.15	983	83		7	
1925.15	1173	123		7	
1133.15		123		7	
		103		7	
		33		7	
		73		7	
		203		7	
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